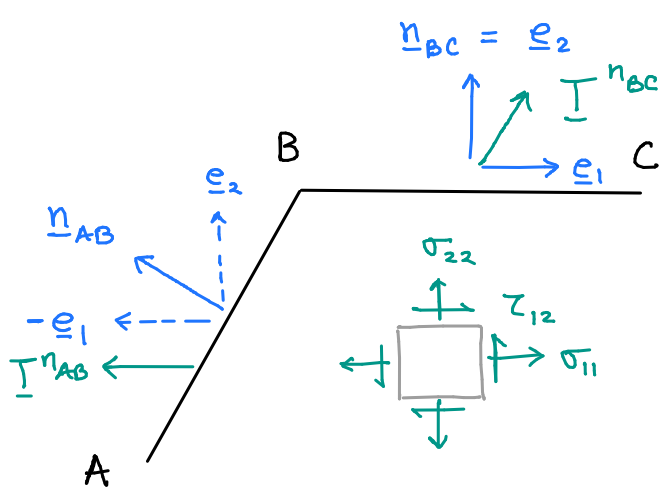


17



On plane BC

$$\begin{bmatrix} T_1^{n_{BC}} \\ T_2^{n_{BC}} \end{bmatrix} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{bmatrix} T_1^{n_{BC}} \\ T_2^{n_{BC}} \end{bmatrix}$$

$$\begin{aligned} T_1^{n_{BC}} &= \underline{T}^{n_{BC}} \cdot \underline{e}_1 \\ &= 20 \cos 40^\circ \\ &= 15.32 \text{ MPa} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} T_2^{n_{BC}} &= \underline{T}^{n_{BC}} \cdot \underline{e}_2 \\ &= 20 \sin 40^\circ \\ &= 12.86 \text{ MPa} \quad \left(\frac{1}{2}\right) \end{aligned}$$

On plane AB

$$\begin{bmatrix} T_1^{n_{AB}} \\ T_2^{n_{AB}} \end{bmatrix} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{bmatrix} T_1^{n_{AB}} \\ T_2^{n_{AB}} \end{bmatrix}$$

$$\begin{aligned} T_1^{n_{AB}} &= \underline{T}^{n_{AB}} \cdot \underline{e}_1 \\ &= 20 \cos 180^\circ \\ &= -20 \text{ MPa} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} T_2^{n_{AB}} &= \underline{T}^{n_{AB}} \cdot \underline{e}_2 \\ &= 20 \cos 90^\circ \\ &= 0 \quad \left(\frac{1}{2}\right) \end{aligned}$$

For a uniform state of stress $\begin{bmatrix} \underline{\sigma} \end{bmatrix} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{bmatrix} \sigma_{11} & \tau_{12} \\ \tau_{12} & \sigma_{22} \end{bmatrix}$

$$\begin{bmatrix} T_1^{n_{AB}} \\ T_2^{n_{AB}} \end{bmatrix} = \begin{bmatrix} \underline{\sigma} \end{bmatrix} \begin{bmatrix} n_{AB} \end{bmatrix}, \quad \begin{bmatrix} T_1^{n_{BC}} \\ T_2^{n_{BC}} \end{bmatrix} = \begin{bmatrix} \underline{\sigma} \end{bmatrix} \begin{bmatrix} n_{BC} \end{bmatrix}$$

where all vectors and matrices are expressed in $(e_1 - e_2)$ coord. sys

$$\begin{bmatrix} n_{AB} \end{bmatrix} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{bmatrix} -\sin 40^\circ \\ \cos 40^\circ \end{bmatrix}, \quad \begin{bmatrix} n_{BC} \end{bmatrix} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

For $\underline{I}^{n_{BC}}$,

$$\begin{bmatrix} 15.32 \\ 12.86 \end{bmatrix} = \begin{bmatrix} \sigma_{11} & \tau_{12} \\ \tau_{12} & \sigma_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \tau_{12} = 15.32 \text{ MPa} \quad \left(\frac{1}{2}\right)$$

$$\sigma_{22} = 12.86 \text{ MPa} \quad \left(\frac{1}{2}\right)$$

For $\underline{I}^{n_{AB}}$,

$$\begin{bmatrix} -20 \\ 0 \end{bmatrix} = \begin{bmatrix} \sigma_{11} & \tau_{12} \\ \tau_{12} & \sigma_{22} \end{bmatrix} \begin{bmatrix} -\sin 40^\circ \\ \cos 40^\circ \end{bmatrix}$$

$$\Rightarrow \sigma_{11} (-\sin 40^\circ) + \tau_{12} \cos 40^\circ = -20$$

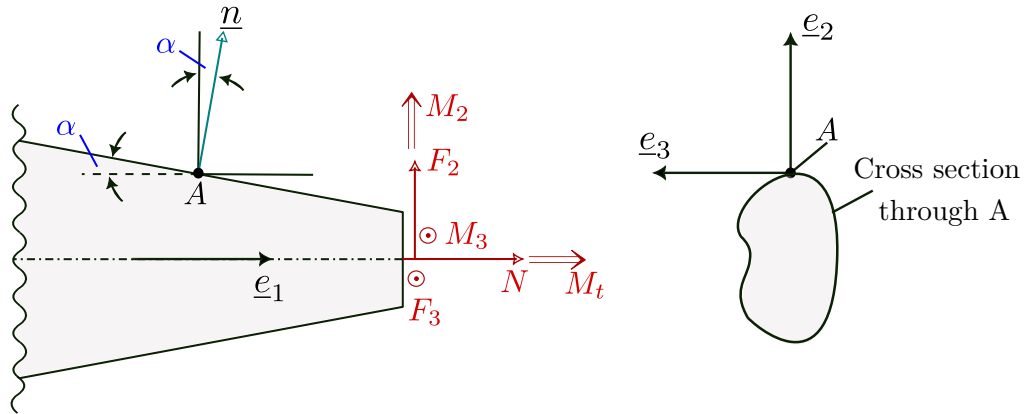
$$\tau_{12} (-\sin 40^\circ) + \sigma_{22} \cos 40^\circ = 0$$

$$\sigma_{11} = \frac{-20 - \tau_{12} \cos 40^\circ}{-\sin 40^\circ} = 49.37 \text{ MPa} \quad \left(\frac{1}{2}\right)$$

Total marks: 20 Total time: 2 hours

Answers without clear motivation/reasoning and handwriting will get zero marks!

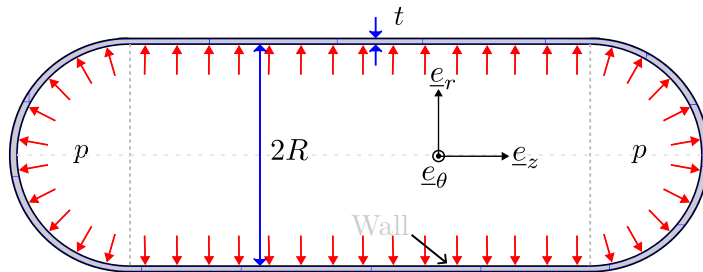
1. [2 marks] At point A of a beam (shown below), the components of the stress tensor expressed in $\underline{e}_1 - \underline{e}_2 - \underline{e}_3$ CSYS are as follows: $\sigma_{11}, \tau_{21}, \tau_{31}$ are non-zero and $\sigma_{22} = \sigma_{33} = \tau_{32} = 0$. $\underline{n}$ is the normal to the lateral surface at A with $\alpha \ll 1$.



Find the error in the satisfaction of the traction-free (i.e., zero traction) boundary condition at A for the beam subjected to bending moment, axial force, and shear force as shown in the diagram.

2. [4 marks] Consider a closed-end **thin-walled** cylindrical shell (like a capsule) with internal radius R and thickness t subjected to a uniform internal pressure p . Show that the principal stresses in the **cylinder wall** (not the end cap walls), in cylindrical coordinates ($\underline{e}_r - \underline{e}_\theta - \underline{e}_z$), are given by:

$$\sigma_{rr} \approx 0, \quad \sigma_{\theta\theta} = \frac{pR}{t}, \quad \sigma_{zz} = \frac{pR}{2t}$$

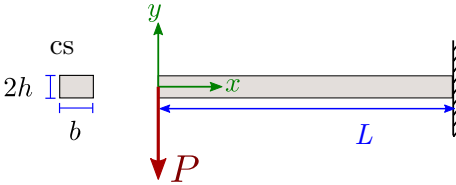


What is the maximum shear stress in the $\theta - z$ plane? Is it equal to the absolute maximum shear stress?

3. [4 marks] The deviatoric stress state is used in theories of failure since it is a measure of those stresses that produce only distortional (and no volume) changes in a body. Consider the state of stress (in MPa):

$$\begin{bmatrix} \underline{\sigma} \end{bmatrix} = \begin{bmatrix} 55 & 0 & 24 \\ 0 & 46 & 0 \\ 24 & 0 & 43 \end{bmatrix}$$

- (a) Determine the deviatoric stress matrix, $\begin{bmatrix} \underline{\sigma}_{\text{dev}} \end{bmatrix}$. (1 mark)
- (b) Compute the stress invariants of the stress tensor $\underline{\sigma}$ and those of deviatoric stress tensor $\underline{\sigma}_{\text{dev}}$. (1 mark)
- (c) Compute the principal directions of the deviatoric stress tensor $\begin{bmatrix} \underline{\sigma}_{\text{dev}} \end{bmatrix}$ and the stress tensor $\begin{bmatrix} \underline{\sigma} \end{bmatrix}$. How are they related?
How are the *absolute* maximum shear stresses related in both cases? (2 marks)
4. [4 marks] Consider a cantilever beam (with one end free and another end fixed) with a vertical load P applied to the free end of the beam. Take origin at the free end. At the fixed end, the displacements and the slope are zero, i.e., $u_x(L, 0) = u_y(L, 0) = \frac{\partial u_y}{\partial x}(L, 0) = 0$. The corresponding strain field at any point in the beam is given by:

$$\begin{aligned} \epsilon_{xx} &= \frac{Pxy}{EI}, & \epsilon_{yy} &= -\nu \epsilon_{xx}, \\ \epsilon_{xz} &= \epsilon_{yz} = \epsilon_{zz} = 0, \\ \epsilon_{xy} &= \frac{P}{4GI}(h^2 - y^2) \end{aligned}$$


ν, E, I, G, L, h, b, P are known constants.

Check if the strain field is compatible or not. Using the given information, find out the displacement fields $u_x(x, y)$ and $u_y(x, y)$.

5. [3 marks] A thin rectangular sheet is strained with uniform strain components: $\epsilon_{xx} = 0.001$, $\epsilon_{yy} = -0.001$, $\gamma_{xy} = 0.001$. All other strain components are zero implying it is a plane strain condition.
- (a) Draw Mohr's circle for this case. (1 mark)
- (b) How much will be the percentage change in the area of the thin sheet? (0.5 mark)
- (c) Which line elements undergo maximum and minimum normal strain? (1 mark)
- (d) What will be the orientation of the two perpendicular lines in the plane of the sheet that undergo maximum angle change between them? (0.5 mark)
6. [3 marks] Show that $\underline{T}^n = \sum_i^3 \underline{T}^i (n \cdot \underline{e}_i) = \sum_i^3 \underline{T}^i (n \cdot \hat{\underline{e}}_i)$, i.e., the formula is independent of what three planes are chosen to determine $\underline{T}^n$!

$$Q1) \quad [\underline{n}] \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = \begin{bmatrix} \sin \alpha \\ \cos \alpha \\ 0 \end{bmatrix} \quad \text{(assuming a unit normal)} \quad \left(\frac{1}{2} \right)$$

Since the surface is assumed to be traction-free (i.e. zero traction boundary), it must satisfy

$$\underline{\underline{\sigma}} \underline{n} = \underline{0} \quad \left(\frac{1}{2} \right)$$

However, for the given components of $[\underline{\underline{\sigma}}] \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix}$, it does not satisfy zero traction.

$$\left(\frac{1}{2} \right) \left\{ \begin{bmatrix} \sigma_{11} & \tau_{21} & \tau_{31} \\ \tau_{21} & \cancel{\sigma_{22}} & \cancel{\tau_{32}} \\ \tau_{31} & \cancel{\tau_{32}} & \cancel{\sigma_{33}} \end{bmatrix} \begin{bmatrix} \sin \alpha \\ \cos \alpha \\ 0 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right.$$

The errors are

$$\begin{aligned} \text{error 1} &= \sigma_{11} \sin \alpha + \tau_{21} \cos \alpha \\ \text{error 2} &= \tau_{21} \sin \alpha \\ \text{error 3} &= \tau_{31} \sin \alpha \end{aligned} \quad \left(\frac{1}{2} \right)$$

2) Generally, the state of stress can vary from point to point.

Therefore, $\sigma_{rr}, \sigma_{\theta\theta}, \sigma_{zz}, \tau_{z\theta}, \tau_{\theta z}, \tau_{rz} \rightarrow f(r, \theta, z)$

- Due to symmetry about z-axis, there would be no variation of any stresses along z-direction and θ -direction

$$\therefore \sigma_{rr}, \sigma_{\theta\theta}, \sigma_{zz}, \tau_{z\theta}, \tau_{\theta z}, \tau_{rz} \rightarrow f(r)$$

$\frac{1}{2}$

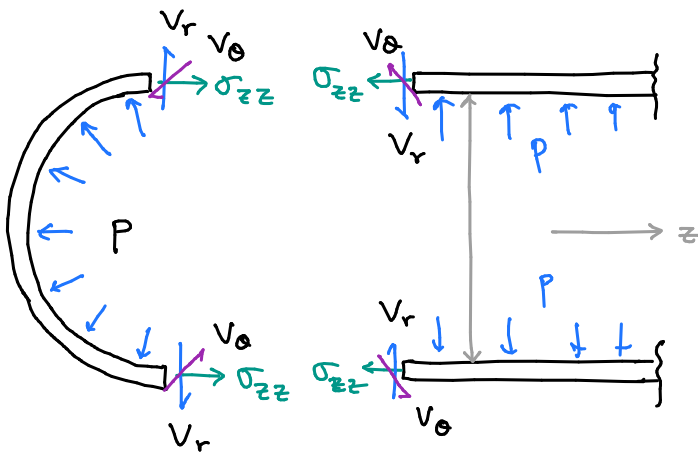
- $\sigma_{rr} = \tau_{rz} = \text{constant}$ (due to negligible variation across in cylindrical wall)

But, $\sigma_{rr}|_{\text{inner wall}} = \tau_{rz} = \sigma_{rr}|_{\text{outer wall}} = \text{constant}$

$\frac{3}{4}$ $\left[\begin{aligned} \therefore \sigma_{rr}|_{\text{outer face}} &= 0 \\ \sigma_{rr}|_{\text{inner face}} &= -P \\ \tau_{rz} &= 0 \end{aligned} \right]$

Wall $\rightarrow$

- At the joint of the end caps to the cylindrical wall,



Using force equilibrium in z-direction:

$$\sum F_z = 0$$

$$\Rightarrow \sigma_{zz} (2\pi r t) = P \pi r^2$$

$$\Rightarrow \sigma_{zz} = \frac{P r}{2t}$$

$\frac{1}{2}$

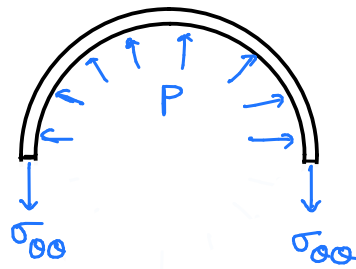
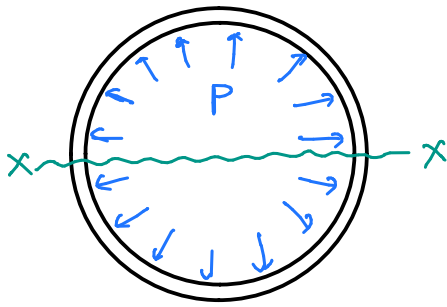
$\frac{1}{2}$ $\left[V_r = 0 \right]$ (else one side should bulge out and other side should bulge in)

$$\therefore \tau_{z\theta} = 0$$

(since there is no twisting moment about z)

$\frac{1}{2}$

Now looking at a cross-section of cylinder wall



$$+\uparrow \sum F_y = 0 \Rightarrow 2 \sigma_{\theta\theta}(t) \cdot (1) = p(2r)(1)$$

$$\Rightarrow \sigma_{\theta\theta} = \frac{pr}{t}$$

There would be no shear, so $\tau_{r\theta} = 0$

Maximum shear stress in θ - z plane $\rightarrow \left| \frac{\gamma_{\theta} - \gamma_z}{2} \right|$

$$= \left| \frac{\sigma_{\theta\theta} - \sigma_{zz}}{2} \right|$$

$$= \frac{\frac{pr}{t} - \frac{pr}{2t}}{2}$$

$$= \frac{pr}{4t} \left\{ \frac{1}{2} \right\}$$

Absolute max. shear stress $\rightarrow \left| \frac{\gamma_{\theta} - \gamma_r}{2} \right| = \left| \frac{\sigma_{\theta\theta} - \sigma_{rr}}{2} \right|$

$$= \frac{pr}{2t} \left\{ \frac{1}{2} \right\}$$

No they are not equal!

$$3) \quad (a) \quad [\underline{\sigma}_{dev}] = \begin{bmatrix} 55 & 0 & 24 \\ 0 & 46 & 0 \\ 24 & 0 & 43 \end{bmatrix} - 48 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 0 & 24 \\ 0 & -2 & 0 \\ 24 & 0 & -5 \end{bmatrix} \quad (1) \quad \frac{55 + 46 + 43}{3}$$

$$(b) \quad I_1(\sigma) = 144 \quad \left\{ \frac{1}{8} \right\}$$

$$I_2(\sigma) = \begin{vmatrix} 55 & 0 \\ 0 & 46 \end{vmatrix} + \begin{vmatrix} 46 & 0 \\ 0 & 43 \end{vmatrix} + \begin{vmatrix} 55 & 24 \\ 24 & 43 \end{vmatrix}$$

$$= 2530 + 1978 + 1789$$

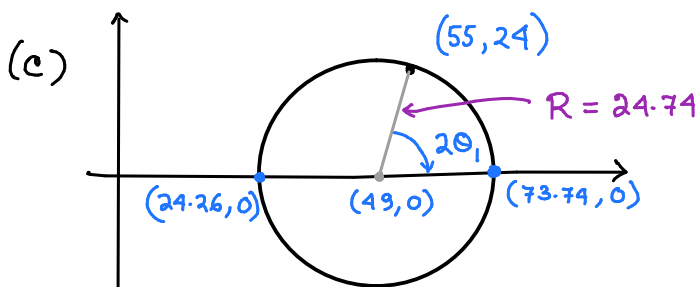
$$= 6297 \quad \left\{ \frac{2}{8} \right\}$$

$$I_3(\sigma) = \det([\underline{\sigma}]) = 82294 \quad \left\{ \frac{1}{8} \right\}$$

$$I_1(\sigma_{dev}) = 0 \quad \left\{ \frac{1}{8} \right\}$$

$$I_2(\sigma_{dev}) = -615 \quad \left\{ \frac{2}{8} \right\}$$

$$I_3(\sigma_{dev}) = 1222 \quad \left\{ \frac{1}{8} \right\}$$



We can draw Mohr's circle for

$$[\underline{\sigma}] = \begin{bmatrix} 55 & 24 \\ 24 & 43 \end{bmatrix}$$

State of stress
 $[\underline{\sigma}]$

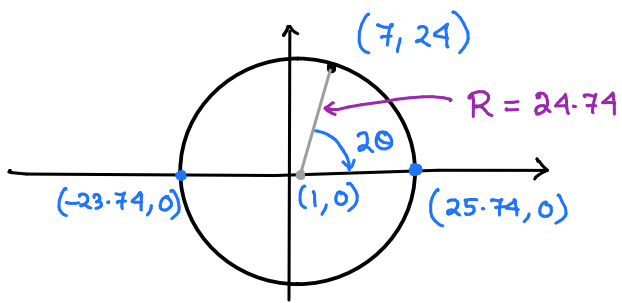
Principal stresses = 73.74, 46, 24.26
of deviatoric state

$\left(\frac{1}{2} \right)$

Angles: $\theta_1 = \frac{1}{2} \tan^{-1} \left(\frac{24}{6} \right)$ anticlockwise from x-axis

$$\theta_2 = \frac{\pi}{2} + \theta_1$$

$\theta_3 =$ along y-axis



We can draw Mohr's circle for

$$[\underline{\sigma}]_{\text{dev}} = \begin{bmatrix} 7 & 24 \\ 24 & -5 \end{bmatrix}$$

Deviatoric

Principal stresses = 25.74, -2, -23.74
of deviatoric state

Angles : $\theta_1 = \frac{1}{2} \tan^{-1} \left(\frac{24}{6} \right)$ anticlockwise from x-axis

$$\theta_2 = \frac{\pi}{2} + \theta_1$$

θ = along y-axis

$\frac{1}{2}$

Principal directions remain the same!

Alternatively,

$$[\underline{n}_1] = \begin{bmatrix} \cos \theta_1 \\ 0 \\ \sin \theta_1 \end{bmatrix}, \quad [\underline{n}_2] = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad [\underline{n}_3] = \begin{bmatrix} \cos \theta_2 \\ 0 \\ \sin \theta_2 \end{bmatrix}$$

(c) Principal stresses for $[\underline{\sigma}] \rightarrow 73.74, 46, 24.26$

$$\begin{aligned} \text{Absolute max shear stress} &= \frac{73.74 - 24.26}{2} \\ &= 24.74 \end{aligned}$$

$\frac{1}{2}$

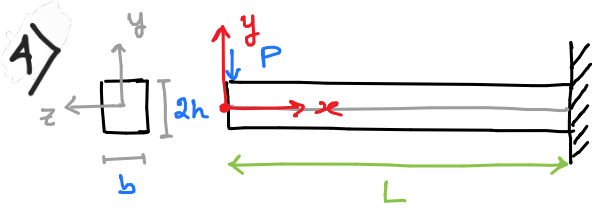
Principal stresses = 25.74, -2, -23.74
of deviatoric state

$$\begin{aligned} \text{Absolute max shear stress} &= \frac{25.74 - (-23.74)}{2} \\ &= 24.74 \end{aligned}$$

$\frac{1}{2}$

They are same!





Given:

$$\epsilon_{xx} = \frac{Pxy}{EI}$$

$$\epsilon_{yy} = -\nu \epsilon_{xx}$$

$$\epsilon_{xz} = \epsilon_{yz} = \epsilon_{zz} = 0$$

$$\epsilon_{xy} = \frac{P}{4GI} (h^2 - y^2)$$

$$u_x|_{(L,0)} = 0$$

$$u_y|_{(L,0)} = 0$$

$$\left. \frac{\partial u_y}{\partial x} \right|_{(L,0)} = 0$$

Integrate the normal strains ϵ_{xx} and ϵ_{yy}

$$\frac{\partial u_x}{\partial x} = \epsilon_{xx} \Rightarrow u_x(x, y) = \frac{Px^2y}{2EI} + g(y) \quad \text{--- (1)}$$

$$\frac{\partial u_y}{\partial y} = -\nu \epsilon_{xx} \Rightarrow u_y(x, y) = -\nu \frac{Px y^2}{2EI} + h(x) \quad \text{--- (2)}$$

From shear strain, we find

$$\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} = 2\epsilon_{xy} = \frac{P}{2GI} (h^2 - y^2)$$

which yields using (1) and (2)

$$\frac{Px^2}{2EI} + \frac{dg}{dy} - \frac{\nu P y^2}{2EI} + \frac{dh}{dx} = \frac{P}{2GI} (h^2 - y^2)$$

Rearranging the terms, we get:

$$\underbrace{\frac{Px^2}{2EI} + \frac{dh(x)}{dx}}_{H(x)} = \underbrace{\frac{P}{2GI} (h^2 - y^2) + \frac{\nu P y^2}{2EI} - \frac{dg(y)}{dy}}_{G(y)}$$

The above equation is of the form $H(x) = G(y)$. Hence, it must be that

$$H(x) = G(y) = C \quad (\text{constant})$$

Therefore, integrating the LHS and RHS, we get

$$\text{LHS: } \frac{Px^2}{2EI} + \frac{dh(x)}{dx} = C \Rightarrow h(x) = Cx - \frac{Px^3}{6EI} + D_1$$

$$\text{RHS: } \frac{P}{2GI} (h^2 - y^2) + \frac{\nu Py^2}{2EI} - \frac{dg(y)}{dy} = C$$

$$\Rightarrow g(y) = -Cy + \frac{P}{2GI} \left(h^2 y - \frac{y^3}{3} \right) + \frac{\nu Py^3}{6EI} + D_2$$

Putting these values in the displacement expressions, we get:

$$u_x(x, y) = \frac{Px^2y}{2EI} + \frac{P}{2GI} \left(h^2 y - \frac{y^3}{3} \right) + \frac{\nu Py^3}{6EI} - Cy + D_2$$

$$u_y(x, y) = -\frac{Px^3}{6EI} - \frac{\nu Pxy^2}{2EI} + Cx + D_1$$

Now, use the displacement boundary conditions

$$u_x|_{(L, 0)} = 0 \Rightarrow D_2 = 0$$

$$u_y|_{(L, 0)} = 0 \Rightarrow D_1 = -CL + \frac{PL^3}{6EI}$$

$$\frac{\partial u_y}{\partial x} \Big|_{(L, 0)} = 0 \Rightarrow -\frac{PL^2}{2EI} + C = 0 \Rightarrow C = \frac{PL^2}{2EI}$$

$$\therefore D_1 = -\frac{PL^3}{3EI}, \quad D_2 = 0, \quad C = \frac{PL^2}{2EI} \quad \left. \vphantom{\frac{PL^3}{3EI}} \right\} \textcircled{1.5}$$

For plane strain case, five strain-compatibility equations are naturally satisfied. You will also find that

$$\star \quad \frac{\partial^2 \epsilon_{11}}{\partial y^2} + \frac{\partial^2 \epsilon_{22}}{\partial x^2} = 2 \frac{\partial^2 \epsilon_{12}}{\partial x \partial y} \quad \text{is also satisfied!} \quad \left. \vphantom{\frac{\partial^2 \epsilon_{11}}{\partial y^2}} \right\} \textcircled{1/2}$$

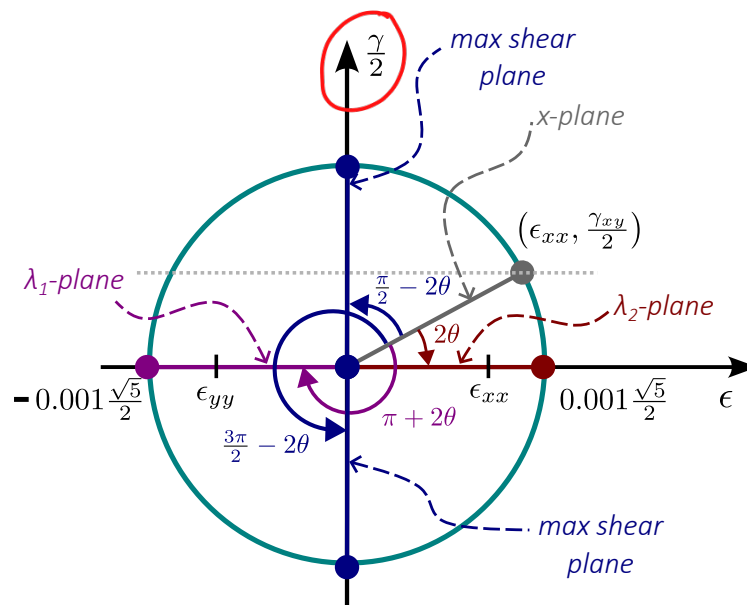
Q57

A thin rectangular sheet is strained with uniform strain components: $\epsilon_{xx} = 0.001$, $\epsilon_{yy} = -0.001$, $\gamma_{xy} = 0.001$. All other strain components are zero implying it is a plane strain condition.

- Draw Mohr's circle for this case.
- How much will be the percentage change in the area of thin sheet?
- Which line elements undergo maximum and minimum longitudinal/normal strain?
- What will be the orientation of the two perpendicular lines in the plane of sheet which undergo maximum angle change between them?

Solution:

$\epsilon_{xx} = 0.001$, $\epsilon_{yy} = -0.001$, $\gamma_{xy} = 0.001$, others are all zero!



- The Mohr's circle is drawn above.

$$\tan 2\theta = \frac{\gamma_{xy}/2}{(\epsilon_{xx} - \epsilon_{yy})/2} = \frac{0.0005}{0.001} = \frac{1}{2}$$

$$\text{Center of the circle} = \frac{\epsilon_{xx} + \epsilon_{yy}}{2} = 0$$

$$\begin{aligned} \text{Radius} &= \sqrt{\left(\frac{\epsilon_{xx} - \epsilon_{yy}}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} \\ &= \sqrt{0.001^2 + \frac{0.001^2}{4}} \\ &= 0.001 \frac{\sqrt{5}}{2} \end{aligned}$$

- Percentage change in area of the sheet = $(\epsilon_{xx} + \epsilon_{yy}) \times 100 = 0\%$

(iii) To find line elements undergoing max. and min. longitudinal strain (principal strains), we use the Mohr's circle. The principal strains are:

- $\frac{\epsilon_{xx} + \epsilon_{yy}}{2} + R = R$: the line element that undergoes maximum longitudinal strain lie along the normal of maximum principal strain plane. Its normal is oriented at

$$2\theta = \tan^{-1} \left(\frac{1}{2} \right) \text{ clockwise from X-axis in Mohr's circle}$$

$$\Rightarrow \theta = \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right) \text{ anti-clockwise from X-axis physically. }] \left(\frac{1}{2} \right)$$

- $\frac{\epsilon_{xx} + \epsilon_{yy}}{2} - R = -R$; the line element that undergoes minimum longitudinal strain lie along the normal of minimum principal strain plane. Its normal is oriented at

$$\pi + 2\theta = \pi + \tan^{-1} \left(\frac{1}{2} \right) \text{ clockwise from X-axis in Mohr's circle}$$

$$\Rightarrow \frac{\pi}{2} + \theta = \frac{\pi}{2} + \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right) \text{ anti-clockwise from X-axis physically. }] \left(\frac{1}{2} \right)$$

(iv) Maximum shear strain will occur between those two line elements which correspond to top and bottom points in the circle. The corresponding directions are:

- First line element oriented along

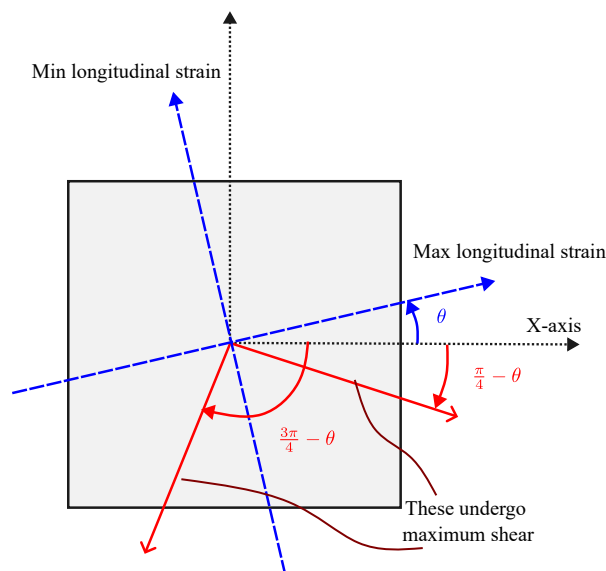
$$\left(\frac{\pi}{2} - 2\theta \right) \text{ anti-clockwise from X-axis in Mohr's circle}$$

$$\Rightarrow \left(\frac{\pi}{4} - \theta \right) \text{ clockwise from X-axis physically }] \left(\frac{1}{4} \right)$$

- Second line element oriented along

$$\left(\frac{3\pi}{2} - 2\theta \right) \text{ anti-clockwise from X-axis in Mohr's circle}$$

$$\Rightarrow \left(\frac{3\pi}{4} - \theta \right) \text{ clockwise from X-axis physically }] \left(\frac{1}{4} \right)$$



Part marks here are calculated based on the Mohr's circle drawn (even if incorrect Mohr's circle)

Q6) This question is from Tutorial 3

$$\underline{e}_j = \sum_i (\underline{e}_j \cdot \hat{\underline{e}}_i) \hat{\underline{e}}_i \quad \} \textcircled{1}$$

$$\underline{T}^n = \sum_{i=1}^3 \underline{T}^{\hat{i}} (\underline{n} \cdot \hat{\underline{e}}_i)$$

Express $\underline{T}^{\hat{i}}$ in terms of traction on planes $\underline{e}_1, \underline{e}_2, \underline{e}_3$

$$\underline{T}^{\hat{i}} = \sum_{j=1}^3 \underline{T}^j (\hat{\underline{e}}_i \cdot \underline{e}_j) \quad \} \textcircled{1/2}$$

Now sub the value of $\underline{T}^{\hat{i}}$ in the expression of $\underline{T}^n$

$$\begin{aligned} \underline{T}^n &= \sum_{i=1}^3 \underline{T}^{\hat{i}} (\underline{n} \cdot \hat{\underline{e}}_i) \\ &= \sum_{i=1}^3 \left(\sum_{j=1}^3 \underline{T}^j (\hat{\underline{e}}_i \cdot \underline{e}_j) \right) (\underline{n} \cdot \hat{\underline{e}}_i) \\ &= \sum_{i=1}^3 \sum_{j=1}^3 \underline{T}^j \underline{n} \cdot ((\hat{\underline{e}}_i \cdot \underline{e}_j) \hat{\underline{e}}_i) \\ &= \sum_{j=1}^3 \underline{T}^j \underline{n} \cdot \left(\underbrace{\sum_{i=1}^3 (\underline{e}_j \cdot \hat{\underline{e}}_i) \hat{\underline{e}}_i}_{= \underline{e}_j} \right) \\ &= \sum_{j=1}^3 \underline{T}^j (\underline{n} \cdot \underline{e}_j) \\ &= \sum_{i=1}^3 \underline{T}^i (\underline{n} \cdot \underline{e}_i) \end{aligned} \quad \} \textcircled{1/2}$$

①